



Installation Instruction

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Assembly Instructions for Cummins® Off-Highway Engines

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1. Introduction

1.1 About These Assembly Instructions

The purpose of this document is to provide assembly instructions for Cummins® Off-Highway Engines. These instructions should be followed during the assembly and installation of the engine into the OEM equipment/machine.

OEMs are responsible for complying with the requirements of Cummins® Advisor Installation Requirements (AIRs) which highlight interfaces, installation requirements, and operating limits in order to achieve "Approved Installation" status. An Approved Installation indicates all applicable Cummins® AIRs have been verified/documented jointly between the OEM Powertrain Engineer and the Cummins® Application Engineering Team. All pertinent AIR information necessary for engine installation will be contained in OEM Manuals.

The production engines Cummins Inc. provide vary in option content. Engine Serial Number (ESN) specific information should be evaluated to ensure that appropriate installation requirements and manuals are referenced during installation. Should further assistance be required, please contact your OEM Powertrain Engineer or authorized Cummins® distributor.

References can be made using Cummins® ESN specific literature such as the Operation and Maintenance Manual, Service Manual, Fault Code Troubleshooting Manual and Fluids for Cummins® Products Service Manual. Service literature is accessible using the following websites:

- [https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html)
([https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html))
- <https://parts.cummins.com/home> (<https://parts.cummins.com/home>)

Region and language settings are determined by web browser when using:

- [https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html)
([https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html))

VPN users may need to update region and language preferences in upper right corner.

In line with Cummins® policy of continuous development and improvement, these documents may be updated over time. Users should ensure that before commencing any work, they have reviewed the latest information available.

These assembly instructions also contain a description of the conditions that **must** be satisfied, so as to **not** compromise health and safety.

Engine installer **must** follow all General Safety Instructions in Operation and Maintenance Manual located at:

- [https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html)

(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)

Many accidents are caused by failure to follow fundamental rules and precautions.

It is the engine installer's responsibility to employ competent people to carry out any installation work.

1.1.1 Intended Use

Cummins® Off-Highway Engines received by OEMs are considered Partly Completed Machinery (PCM).

The PCM in this case is an internal combustion diesel engine. A diesel engine is defined as an internal combustion engine in which ignition of diesel fuel is caused by the elevated temperature of the air in the cylinder due to mechanical compression.

The PCM will **not** function until the OEM integrates into equipment and all installation requirements are met and deemed an "Approved Installation" by Cummins®. The PCM will arrive at engine installer's facility secured to material handling hardware such as a skid or enclosure.

1.2 Warning, Caution, and Note Styles Used in Manuals

The following safety styles and symbols found throughout manuals indicate potentially hazardous conditions to the operator, service personnel, or equipment.

WARNING

Serious personal injury or extensive property damage can result if the warning instructions are not followed.

CAUTION

Minor personal injury can result or a part, and assembly, or the engine can be damaged if the caution instructions are not followed.

1.3 Safety Precautions

Read these instructions completely and become familiar with safety warnings, cautions, and procedures before starting the installation.

Only authorized and qualified maintenance technicians, who are familiar with the equipment and its operation, should carry out maintenance.

Before carrying out any maintenance work, become familiar with engine safety by reviewing all the documentation supplied with the engine.

Before carrying out any maintenance work, lock the machine and/or engine out of service for safe working.

To reduce the possibility of personal injury, personal protective equipment (PPE) is recommended to be utilized. Always meet on-site safety regulations for required PPE. Proper maintenance of safety equipment **must** be practiced. Integrity of safety equipment **must** be checked to verify

equipment functionality is maintained.

Incorrect Installation:

⚠ WARNING ⚠

Incorrect installation, service, or parts replacement can result in severe personal injury, and/or equipment damage. Service personnel must be trained and experienced to perform electrical and mechanical component installations.

Fall Hazard:

⚠ WARNING ⚠

Falls can result in severe personal injury. Make sure that suitable equipment for performing tasks at height are used in accordance with local guidelines and legislation.

Lifting:

⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and the OEM powertrain/upfitted components as an assembly without causing personal injury.

⚠ WARNING ⚠

For components that weigh 23 kg [50 lb] or more use a hoist or get assistance to lift to avoid possibility of personal injury.

Sharp Edges and Sharp Points:

⚠ CAUTION ⚠

Projecting corners/parts may cause cuts, abrasions and other personal injury. Use PPE. Be aware of sharp edges and corners/sharp points. Cover/protect them.

Fluid Exposure:

⚠ CAUTION ⚠

For proper handling of various substances, refer to the manufacturer's safety data sheet. PPE must be utilized as required.

Slip Hazard:

⚠ CAUTION ⚠

Spilled coolant, fuels and lubricating fluids are a slip hazard and can cause severe personal injury. Clean up any spilled fluids before operating the engine.

1.3.1 Installation Tools

All tools used in the installation shall be suitable for the task, in good condition and free from damage.

Only trained and experienced technicians shall install the product.

Engine installers should reference OEM Manuals for tools required.

Cummins® Service tools that are specific to the ESN and are contained in the Service Manual at:

- <https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>
(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)

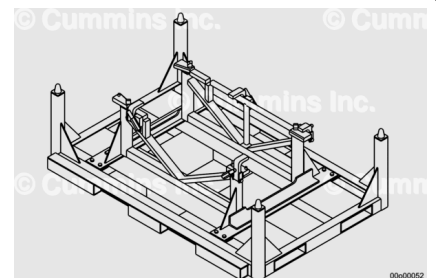
1.3.2 Site Safety

Follow all local and national site safety procedures for installation.

If an incident occurs, alert others and follow the specific site safety processes.

1.4 Transit and Material Handling

- Engine installers **must** take the necessary precautions during moving and handling of the engine while attached to material handling skid.
 - **Always** use forklift slots to pick up material handling skid in safe manner.
 - Verify **only** forklift trained and authorized personnel are moving material handling skids.
 - Verify forklift capacity is **not** exceeded.
 - Verify material handling skid is secured during transit.
 - X15 Example Skid shown.



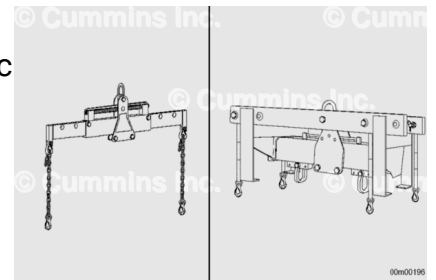
1.5 Lifting

- Engine installers **must** take the necessary precautions when lifting the engine.
 - Clear the area before lifting. Verify nothing is under lifted hardware.
 - **Always** focus your attention on the lift.
 - **Always** verify lifting devices are in working order.
 - **Always** verify lifting device capacity is **not** exceeded.
- The mass of the engine is listed in Operation and Maintenance Manual at:
 - <https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>

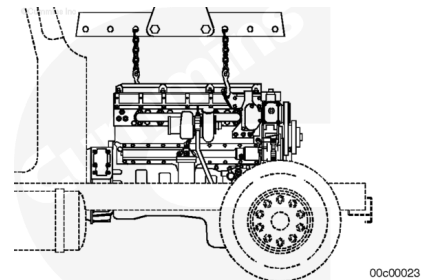
(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)

- The mass stated does **not** include packing materials.
- The mass stated does **not** include OEM upfitted components.

- The correct lifting device **must** be used and attached to the designated lifting brace
 - Lifting devices should be capable of lifting the necessary mass and accommodating the center of gravity.
 - Cummins® Service tools that meet these requirements are specific to the ESN and are contained in the Service Manual at:
 - <https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>
(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)
 - Example Cummins® Service Tools shown.



- All lifting operations **must** be carried out by qualified personnel and using suitably rated and tested equipment in accordance with all applicable safety procedures.
 - Example Cummins® Service Tool in practice shown.



1.6 Protection From Hot Surfaces

The engine may be supplied with heat shielding hardware to guard against hot surfaces. Heat shielding provided with the engine **must not** be modified.

Other surfaces that may get hot include but are **not** limited to cylinder block, cylinder head, exhaust manifold and turbocharger components.

The engine installer **must** make sure that protections from hot surfaces comply with local and national regulations.

1.7 Protection From Moving Parts

If guards provided with the engine for moving parts are disturbed or modified during installation, the engine installer **must** comply with local and national regulations.

1.8 Protection From Noise

High Noise Level:



Engines in operation emit noise, which can cause hearing damage. Wear appropriate ear protection at all times.

After installation the engine can emit noise during operation. The engine installer is responsible for assessing any hazard that may result from noise emissions and for providing warnings and hearing protection advice.

The engine installer is responsible for making sure conditions comply with local and national regulations.

1.9 Protection From Vibration

The equipment produces vibration during operation. The engine installer is responsible for assessing the effects of vibration on the installation design and for including appropriate vibration considerations within the machinery design or installation.

1.10 Control System

The engine is supplied without a control system. The engine installer **must** make sure that a suitable control and wiring system is fitted to control and protect the equipment in accordance with local and national regulations.

1.11 Stability

The engine installer is responsible for making sure that the equipment installation complies with local and national regulations, and is stable under all foreseeable conditions, such as:

- Load
- Overload
- Fault Conditions
- Conditions during Transit
- Positioning

All installation, commissioning and operation **must** be completed in accordance with local, national and regional legislation.

1.12 Specifications

The specifications of the engine are listed by ESN in Operation and Maintenance Manual, Service Manual, and Fluids for Cummins® Products Service Manual at:

- [https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html)
([https://quickserve.cummins.com/authenticate?
token=RVVNUg==&target=/integrations/regulations/index.html](https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html))

1.13 How to Obtain Service

When a product requires service, contact the nearest Cummins® service provider. To locate the distributor, go to

- <https://www.cummins.com/en-eu/customer-assistance> (<https://www.cummins.com/en-eu/customer-assistance>)

Select Location on homepage to find a sales or service location. When contacting the service provider, always supply the complete model, specification, and ESN as shown on the nameplate.

2. Parts Information

2.1 Introduction

All parts for Cummins® Engines can be found by ESN at:

- <https://parts.cummins.com/home> (<https://parts.cummins.com/home>)

Contact your authorized distributor with any discrepancies.

3. Maintenance

3.1 Introduction

Engine maintenance and service information is available by ESN in the manuals listed below at:

- <https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>
(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)
 - Operation and Maintenance Manual
 - Service Manual
 - Fluids for Cummins® Products Service Manual
 - Fault Code Troubleshooting Manual

3.2 Maintenance Safety

Engine installer **must** refer to Operation and Maintenance Manual using the following website for General and ESN specific safety information.

- <https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>
(<https://quickserve.cummins.com/authenticate?token=RVVNUg==&target=/integrations/regulations/index.html>)

Document History

Date	Details
2026-6-25	Module Created
2026-6-30	Added graphic of X15 Skid.
2026-7-1	Minor updates made to formatting.
2026-7-2	Fixing a few minor issues. None of the actual text contents have been changed.

Date	Details
2026-7-6	Updated website URLs to ensure they navigate to proper page.
2026-8-6	Updated website URLs to ensure they navigate to the proper page. No content changes made to service bulletin.

Last Modified: 07-Aug-2026
